Tools — Git

Local and remote git operations. Remote operations (`git_push`, destructive resets) follow the Version Control Safety rules: never push unless the user explicitly asks; never rewind files with `git_checkout`.

Tool	Description
`git_add`	Stage files for commit.
`git_branch`	List local and remote branches.
`git_checkout`	Switch branch or restore files.
`git_cherry_pick`	Apply changes from existing commits.
`git_clone`	Clone a repository.
`git_commit`	Create a commit from staged changes.
`git_diff`	Show working tree, staged, or commit diff.
`git_fetch`	Fetch from remote without merging.
`git_log`	Show commit history.
`git_merge`	Merge another branch.
`git_pull`	Fetch and integrate from remote.
`git_push`	Push branches and tags to remote.
`git_remote`	List, add, remove, update remotes.
`git_reset`	Unstage or reset to a commit.
`git_show`	Show commit, tag, or object details.
`git_stash`	Stash and unstash changes.
`git_status`	Show working tree status.
`git_tag`	List, create, show, or delete tags.

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git_add

Stage files for the next commit. At least one of `paths`, `all`, or `update` must be provided.

Arguments

Argument	Type	Required	Description	Typical value
`paths`	array	one of three	Specific files or directories to stage	`["src/main.rs"]`
`all`	boolean	one of three	Stage all changes (`git add -A`)	`true`
`update`	boolean	one of three	Stage changes to tracked files only (`git add -u`)	`true`

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git_branch

List local and remote branches, including the checked-out branch.

Arguments

Argument	Type	Required	Description	Typical value
`all`	boolean	no	Include remote-tracking branches	`true`
`format`	enum	no	`short` (terse names) or `verbose` (with upstream tracking info)	`"short"`

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git_checkout

Switch branches, or restore working-tree files. Provide exactly one mode. **Do not use to rewind files** — it risks losing work.

Arguments

Argument	Type	Required	Description
`branch`	string	branch mode	Branch to switch to
`create_branch`	boolean	no	Create and switch (`git checkout -b`)
`paths`	array	restore mode	Files/directories to restore
`source`	string	no	Ref to restore from (default `HEAD`; only with `paths`)

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git_cherry_pick

Apply changes from existing commits onto the current branch.

Arguments

Argument	Type	Required	Description
`commits`	array	yes	Commit hashes or refs to cherry-pick (first must be valid)
`no_commit`	boolean	no	Apply to working tree/index without committing (`-n`)

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git_clone

Clone a repository into a new directory inside the working directory.

Arguments

Argument	Type	Required	Description	Typical value
`url`	string	yes	Repository URL	`"https://github.com/user/repo.git"`
`directory`	string	no	Target folder name (default: inferred from URL)	—
`branch`	string	no	Branch to check out (`--branch`)	`"main"`
`depth`	integer	no	Shallow clone depth (`--depth`)	`1`
`bare`	boolean	no	Create a bare repository	`false`

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git_commit

Create a commit from staged changes.

Arguments

Argument	Type	Required	Description	Typical value
`message`	string	yes	Commit message	`"Fix config loader panic"`
`all`	boolean	no	Also stage modified tracked files (`-a`)	`false`
`amend`	boolean	no	Amend the previous commit	`false`
`no_verify`	boolean	no	Bypass pre-commit hooks	`false`

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git_diff

Show differences for the working tree, the staged index, or a commit.

Arguments

Argument	Type	Required	Description	Typical value
`target`	string	no	`working` (default), `staged`/`cached`, or a commit ref	`"staged"`

Argument	Type	Required	Description	Typical value
`path`	string	no	Limit to one file or directory	`"src/cli.rs"`
`stat`	boolean	no	Compact change summary only	`true`

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git_fetch

Fetch updates from a remote without merging.

Arguments

Argument	Type	Required	Description
`remote`	string	no	Remote name (default `origin`)
`branch`	string	no	Specific branch or ref
`prune`	boolean	no	Remove stale remote-tracking branches
`all`	boolean	no	Fetch every remote (ignores `remote`/`branch`)

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git_log

Show commit history for the current branch or a ref.

Arguments

Argument	Type	Required	Description	Typical value
`limit`	integer	no	Max commits	`20`
`branch`	string	no	Ref to log (default: current branch/HEAD)	`"main"`
`oneline`	boolean	no	Compact one-line format	`true`
`author`	string	no	Filter by author name or email	—
`since`	string	no	Only newer commits	`"2024-01-01"`, `"1.week"`

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git_merge

Merge another branch into the current branch.

Arguments

Argument	Type	Required	Description
`branch`	string	yes	Branch to merge in
`message`	string	no	Custom merge commit message
`no_ff`	boolean	no	Always create a merge commit
`ff_only`	boolean	no	Abort unless fast-forward (mutually exclusive with `no_ff`)
`squash`	boolean	no	Squash incoming changes into one commit

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git_pull

Fetch and integrate changes into the current branch.

Arguments

Argument	Type	Required	Description
`remote`	string	no	Remote name (default `origin`)
`branch`	string	no	Remote branch to pull (default: tracking branch)
`rebase`	boolean	no	Rebase instead of merge

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git_push

Push local branches and tags to a remote. **Only call with explicit user instruction.**

Arguments

Argument	Type	Required	Description
`remote`	string	no	Remote name (default `origin`)
`branch`	string	no	Branch to push (default: current)
`force`	boolean	no	Force with lease (destructive; requires explicit confirmation)
`tags`	boolean	no	Push all tags

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git_remote

List, add, remove, or set URLs for remotes.

Arguments

Argument	Type	Required	Description
`action`	enum	no	`list` (default), `add`, `remove`, `set-url`
`name`	string	yes for non-list	Remote name
`url`	string	yes for `add`/`set-url`	Remote URL

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git_reset

Unstage files or reset the branch to a commit. Choose one mode. **`hard` discards all local working-tree changes.**

Arguments

Argument	Type	Required	Description	Typical value
`paths`	array	unstage mode	Files to unstage	`["src/x.rs"]`
`target`	string	reset mode	Commit hash, tag, or ref (default `HEAD`)	`"HEAD~1"`
`mode`	enum	reset mode	`mixed` (default, keeps changes unstaged), `soft` (keeps staged), `hard`, `keep`	`"mixed"`

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git_show

Show details of a commit, tag, or other git object.

Arguments

Argument	Type	Required	Description	Typical value
`ref`	string	no	Commit hash, tag, or ref (default `HEAD`)	`"abc123"`
`stat`	boolean	no	Include file change statistics (default true)	`true`

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git_stash

Stash and unstash working-tree changes.

Arguments

Argument	Type	Required	Description
`action`	enum	no	`push` (default), `pop`, `apply`, `drop`, `list`, `clear`
`message`	string	no	Message for `push`
`index`	integer	no	Stash index for `pop`/`apply`/`drop` (default 0)

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git_status

Show working-tree status: modified, staged, untracked, conflicted files.

Arguments

Argument	Type	Required	Description
`short`	boolean	no	Compact one-line-per-file format
`branch`	boolean	no	Include current branch name (default true)

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git_tag

List, show, create, or delete tags. Annotated tags require both `name` and `message`. **Do not tag releases unless explicitly asked.**

Arguments

Argument	Type	Required	Description
`action`	enum	no	`list` (default), `show`, `create`, `delete`
`name`	string	yes for show/create/delete	Tag name
`message`	string	annotated create	Tag message
`ref`	string	no	Target commit or ref (default HEAD)

Example:

```
git_status
git_add paths=["src/main.rs"]
git_commit message="Add greet subcommand"
```